PSYC Unified Field-Oriented Control Platform

Technical Reference Manual & Hardware Architecture

Hardware Platform:	**Indus-Drive 60V**
Hardware Revision:	v1.0 (Industrial)
Core Architecture:	STM32G431 + DRV8353RHRGZT

Psyc Aerospace and Defence Industries Pvt.ltd

May 20, 2026

Contents

CHAPTER 1

Executive Summary & Introduction

1.1 Project Objective and Scope

The Indus-Drive 60V is a high-performance industrial-grade Brushless DC (BLDC) motor controller designed specifically for Field-Oriented Control (FOC). Unlike standard hobby Electronic Speed Controllers (ESCs) that rely on trapezoidal commutation, this controller is engineered for aerospace and advanced robotics applications.

The platform integrates:

- Advanced telemetry
- Blackbox data logging
- Active dynamic braking
- Deterministic real-time control

1.2 Unified FOC Controller Philosophy

Following internal engineering standards, this board is classified as a:

Reconfigurable Vector-Controlled PMSM Drive

The architecture is intentionally reusable across multiple electromechanical systems through firmware configuration rather than hardware redesign.

Adjustable parameters include:

- Motor constants
- Control loop gains
- Encoder scaling

Figure 1.1: Gimbals

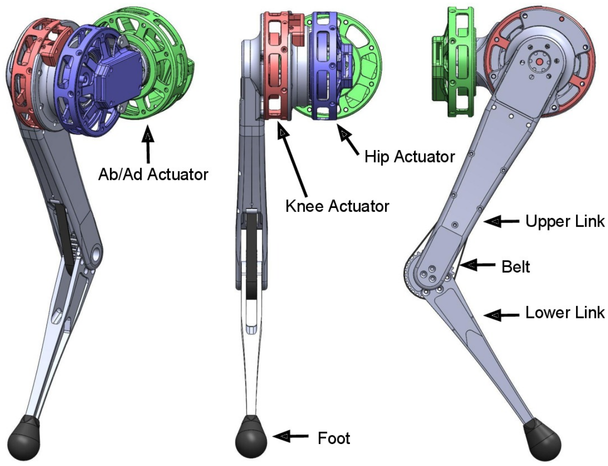

Figure 1.2: Robotic Joints

1.3 Primary Application Domains

The controller supports four validated PSYC motor classes:

Class	KV Range	Applications
PSYC-Torque	30–120	Robotic joints, exoskeletons
PSYC-Precision	120–350	Stabilized gimbals, EO/IR payloads
PSYC-Actuation	350–700	Rotorcraft cyclic pitch control
PSYC-Velocity	700+	UAV propulsion, high-speed fans

1.4 The Case for Field-Oriented Control (FOC)

To justify the architectural complexity of the Indus-Drive 60V, it is critical to understand the limitations of existing commercial off-the-shelf (COTS) alternatives. The industry currently faces a dichotomy between power and precision.

Figure 1.3: UAV Propulsion

Standard ESCs vs. Precision Actuators

Standard drone Electronic Speed Controllers (ESCs) utilize **Trapezoidal (Six-Step) Commutation**. While they can push massive current (50A+), they are essentially "blind." They guess the rotor position using Back-EMF, resulting in severe torque ripple, acoustic screeching, and an inability to operate at low speeds or hold a static load. Conversely, precision robotic actuators utilize FOC for buttery-smooth motion but often lack the hardware capability to handle 60V/50A continuous aerospace loads.

The FOC Advantage

The Indus-Drive 60V bridges this gap. By utilizing a high-resolution absolute magnetic encoder and Space Vector PWM (SVPWM), the board perfectly synchronizes the stator's magnetic field to remain exactly 90 degrees ahead of the rotor.

Parameter	**Standard Drone ESC**	**Indus-Drive 60V (FOC)**
Commutation	6-Step Trapezoidal (Blocky)	Continuous Sinusoidal (SVPWM)
Position Sensing	Sensorless (Back-EMF Guessing)	Absolute Magnetic Encoder (MA732)
Low-Speed Torque	Terrible (Stutters and stalls)	Maximum torque from 0 RPM
Acoustic Noise	High (Mechanical vibration)	Near Silent
Efficiency	Wastes energy on incorrect vectors	100% optimized vector alignment

Table 1.2: Comparison of traditional ESCs versus the Indus-Drive FOC architecture.

CHAPTER 2

System Architecture & Hardware Design

2.1 High-Level Logical Architecture

The system is logically divided into three primary functional blocks: The compute core (Brain), the power stage (Muscle), and the feedback network (Eyes). The closed-loop nature of this architecture ensures deterministic execution of the control loops.

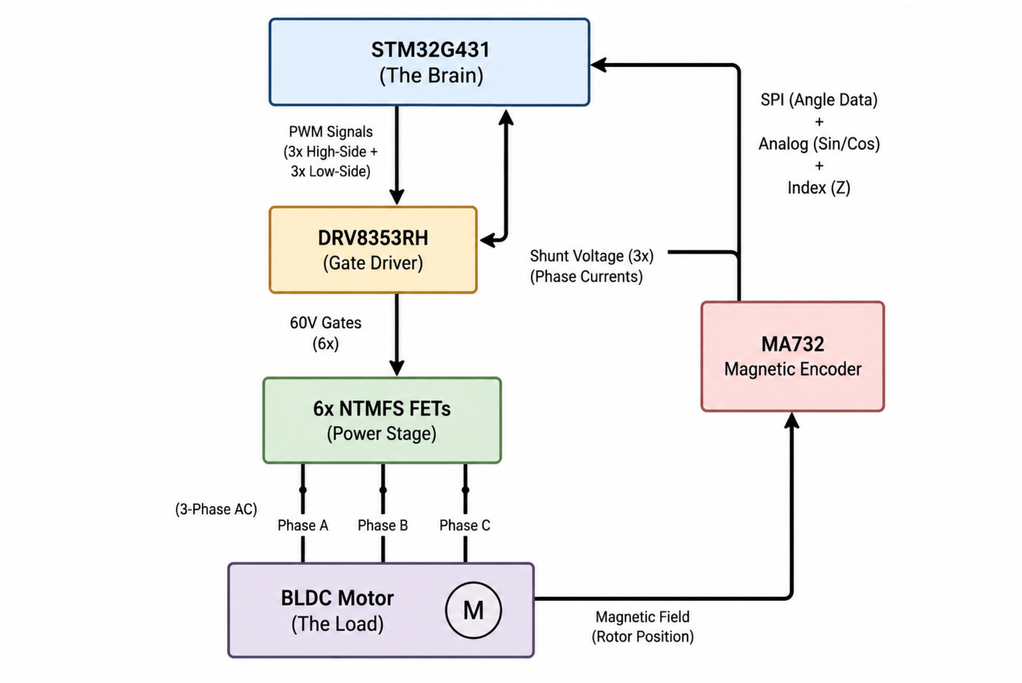

Figure 2.1: High-Level Logical Block Diagram of the Indus-Drive 60V.

2.2 PCB Specifications & 4-Layer Stackup

To achieve a continuous 50A current rating within a heavily constrained **5cm** $\times$ **5cm** footprint, the PCB relies on a meticulously designed 4-layer stackup to manage both high-power thermal dissipation and high-frequency signal integrity.

- **Layer 1 (Top Signal & High Current):** Contains the primary components, MCU, and heavy copper pours for the motor phases (Phase A, B, C) and DC input. Switching nodes are strictly quarantined to the board's perimeter.
- **Layer 2 (Internal GND Plane):** A completely unbroken, solid ground plane. This provides a zero-impedance return path for the high-speed SPI signals and acts as a massive EMI shield between the power stage and the logic layers.
- **Layer 3 (Internal VCC Plane):** Dedicated power distribution plane (3.3V, 5V, and V-Bus routing) to minimize parasitic inductance and ensure stable ADC reference voltages.
- **Layer 4 (Bottom Signal):** Hosts the MA732 Magnetic Encoder dead-center for perfect alignment with the motor shaft magnet, forming an isolated "quiet island" for sensitive analog routing.

2.3 Microcontroller Unit: STM32G431

The computational core is the STM32G431CBT6. Running at **170 MHz** on an ARM Cortex-M4, it provides deterministic execution speed for control loops exceeding 20 kHz.

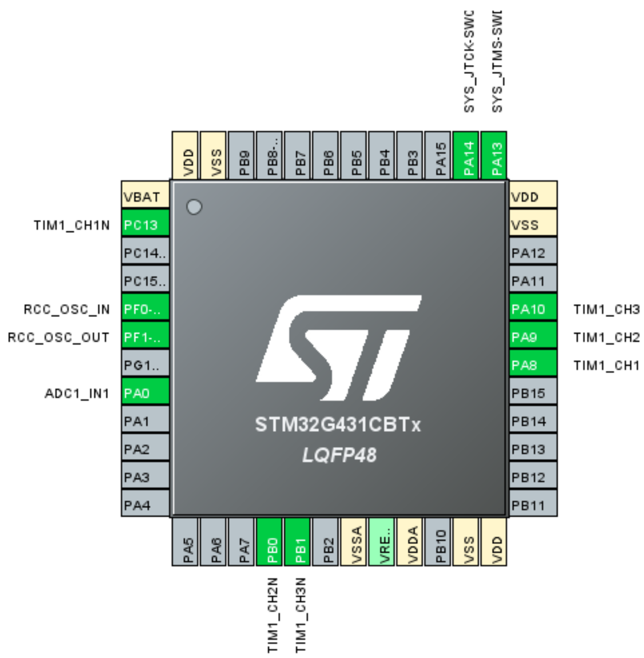

Figure 2.2: STM32G431 Pinout Mapping. Note the grouping of advanced TIM1 PWM outputs (PA8, PA9, PA10) and synchronized ADC inputs (PA0).

CORDIC Accelerator

FOC algorithms rely on trigonometric calculations. The hardware CORDIC coprocessor executes these in only a few clock cycles, allowing concurrent execution of CAN FD communications, Blackbox logging, and real-time motor control loops.

2.4 Smart Gate Driver: DRV8353RHRGZT

The gate driver translates 3.3V logic signals into high-current gate drive signals required to switch 60V MOSFETs efficiently. It features three current sense amplifiers, a hardware overcurrent comparator, and an integrated buck regulator (50.4V $\rightarrow$ 5V logic rail). The design deliberately uses the **hardware-configured variant** rather than SPI to ensure EMI immunity and deterministic, fail-safe protection.

2.5 Power MOSFETs and Voltage Headroom Trade-offs

The power stage uses six NTMFS5C628NL MOSFETs rated for 60V and 2.4 mΩ $R_{DS(on)}$. A 12S LiPo battery peaks at **50.4 V**, leaving only 9.6V of safety margin. Higher voltage MOSFETs (e.g., 80V) would significantly increase $R_{DS(on)}$, doubling conduction losses and causing catastrophic thermal buildup at 50A. This risk is mitigated using 100V ceramic bypass capacitors and an active rheostatic braking circuit.

CHAPTER 3

BLDC Motor Theory & Physics

3.1 Stator and Rotor Fundamentals

A BLDC motor consists of two primary electromechanical components:

- The Stator: The stationary ring consisting of laminated steel sheets with slots that hold copper windings.
- The Rotor: The moving component carrying permanent rare-earth magnets.

3.2 Slot and Pole Combinations

The relationship between the number of stator slots (N) and rotor poles (P) dictates the motor's mechanical performance. Pole Pairs are calculated as $P/2$. This is a critical variable in firmware, as it maps the mechanical angle of the motor to the electrical angle required for FOC commutation.

CHAPTER 4

Field-Oriented Control Implementation

4.1 The FOC Execution Flow

The fundamental mathematical operation of FOC is to measure the chaotic 3-phase AC currents and mathematically flatten them into two easily controllable DC variables.

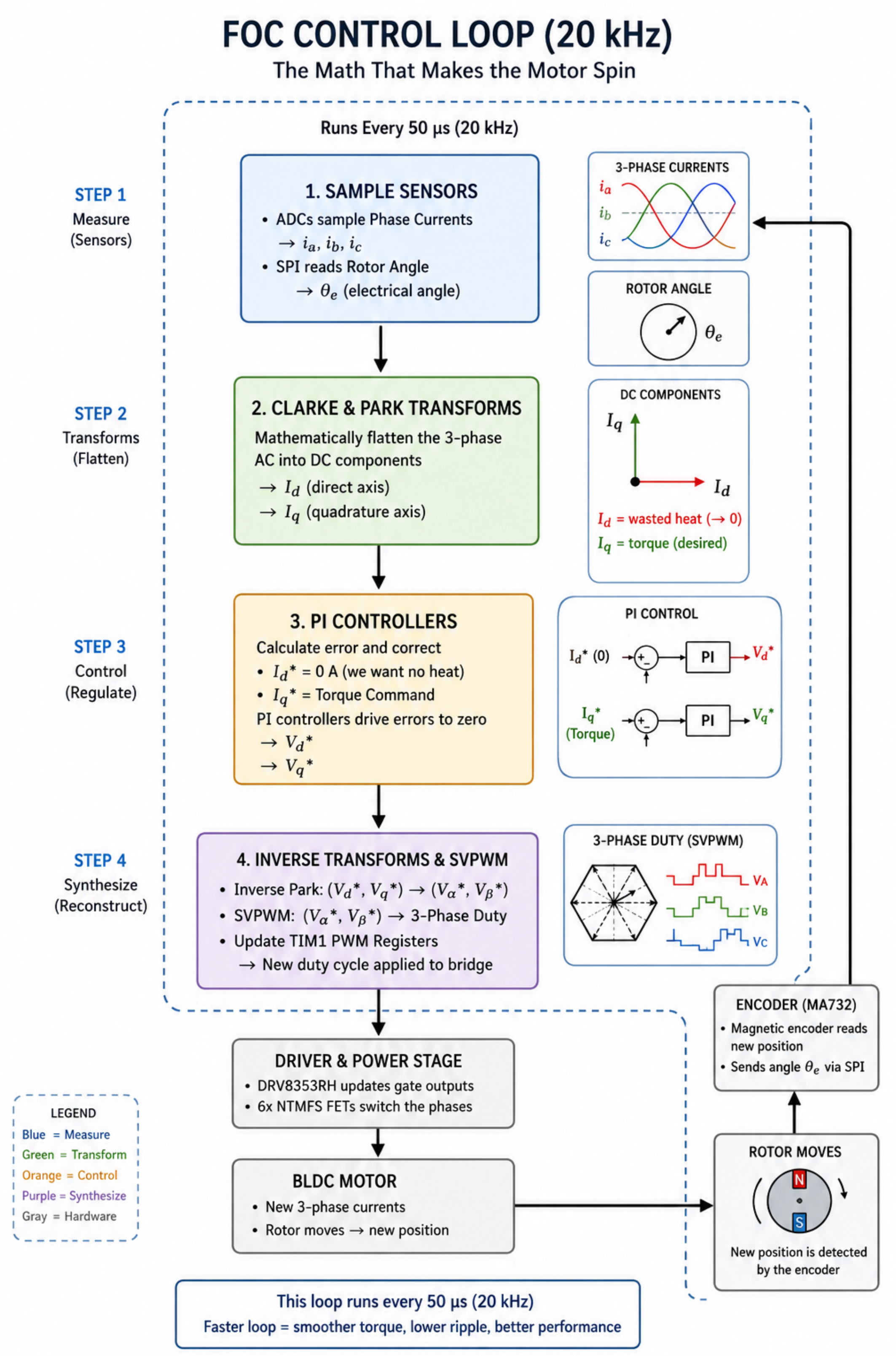

Figure 4.1: The 20 kHz Field-Oriented Control Loop. Illustrating the transformation from physical sensors to mathematical correction, and back to physical PWM synthesis.

4.2 Control Loops and Algorithms

The firmware relies on a cascaded multi-loop architecture. The inner loops execute at extremely high speeds to manage electrical dynamics, while the outer loops manage slower mechanical dynamics.

Control Loop	Frequency	Primary Function
Current Loop (FOC)	20 kHz – 40 kHz	Executes Clarke/Park transforms, PI control for I_d/I_q, and SVPWM synthesis.
Velocity Loop	2 kHz	Calculates error between target RPM and actual RPM, outputs an I_q (Torque) command.
Position Loop	200 Hz	Calculates error between target angle and actual shaft angle, outputs a target Velocity.

Table 4.1: Cascaded Control Loop Frequencies.

4.3 Mathematics of FOC

Step 1: The Clarke Transform (Converts 3-phase a, b, c to 2-phase orthogonal α, β):

$$\begin{aligned} i_\alpha &= i_a, \\ i_\beta &= \frac{1}{\sqrt{3}}(i_b - i_c) \end{aligned} \tag{4.1}$$

Step 2: The Park Transform (Rotates the α/β frame to align with the spinning rotor, using the encoder angle θ):

$$\begin{aligned} i_d &= i_\alpha \cos(\theta) + i_\beta \sin(\theta), \\ i_q &= -i_\alpha \sin(\theta) + i_\beta \cos(\theta) \end{aligned} \tag{4.2}$$

4.4 Critical Firmware Parameters

To achieve absolute stability, the firmware's PI controllers must be mathematically tuned to the exact physical constants of the attached motor using Pole-Zero Cancellation.

Parameter	Symbol	FOC System Function
Stator Resistance	R_s	Used to calculate the Integral Gain (K_i) of the current loop.
Stator Inductance	L_d, L_q	Used to calculate the Proportional Gain (K_p) of the current loop.
Pole Pairs	$P/2$	Scales the mechanical encoder angle into the electrical angle θ_e for transforms.
Flux Linkage	λ_m	Required for field-weakening algorithms and back-EMF observation.

Table 4.2: Key Physical Parameters for FOC Firmware Tuning.

4.5 Hardware Filtering

For the FOC mathematics to work, the STM32 must receive flawless current readings. A strict hardware low-pass filter is implemented on the SOA, SOB, and SOC lines from the DRV8353. With $R = 56\Omega$ and $C = 2.2$ nF, the cutoff frequency (f_c) is **1.29** MHz. This aggressively filters out RF ringing from the MOSFET switching edges without introducing phase delay to the true 20kHz motor current waveform.

CHAPTER 5

Power Stage & Thermal Budgeting

5.1 System Power Limits

The Indus-Drive 60V operates under the following primary power domain specifications:

- Maximum Input Voltage: $12S$ LiPo = 12 cells $\times$ $4.2V$ = $\mathbf{50.4V}$ MAX
- Continuous Phase Current: **50A**
- Peak power capacity: $P = V \times I = 50.4V \times 50A = \mathbf{2,520}$ Watts

5.2 DC Link Capacitance

1. Bulk Electrolytic Capacitors (3x 150uF, 80V) Bulk capacitors act as local energy reservoirs to handle massive ripple currents generated by PWM switching and regenerative braking. Selecting 80V capacitors yields a 58% safety margin over the 50.4V bus. Placing three in parallel divides the Equivalent Series Resistance (ESR) by 3, preventing premature electrolyte dry-out.

2. Ceramic Bypass Capacitors (1210 Package, 100V) To prevent the $L \times (di/dt)$ inductive spikes generated during 20ns switching transients from destroying the MOSFETs, 100V ceramic capacitors with near-zero ESL are placed within 2 mm of the switching nodes to instantly clamp lethal transients.

CHAPTER 6

Subsystems and Interfaces

6.1 CAN Bus Architecture and Telemetry

The controller utilizes a TJA1051T-3 transceiver (U6) for robust CAN FD communication. To facilitate network scalability without Y-splitters, the board features dual 2-pin CAN headers (J3 and J6) allowing for seamless daisy-chaining to a central flight computer (e.g., Pixhawk).

- **Split Termination Network:** To enhance common-mode noise rejection, the termination utilizes a split network consisting of two 62Ω resistors (R43, R44) and a 4.7nF capacitor (C50) to ground.
- **Termination Switch:** A dual-pole SMD switch (SW1) optionally engages this 120Ω equivalent termination across CAN_H and CAN_L. Only the physical end-nodes of the CAN bus chain should have this enabled to prevent signal reflection.

6.2 SPI Bus Arbitration: Encoders and Flash Memory

FOC requires absolute rotor position, and industrial applications demand robust data logging. The board utilizes a shared SPI bus architecture to manage both requirements efficiently.

- **Blackbox Flash Memory (U2):** A 16MB W25Q128JVP SPI NOR Flash is integrated for high-speed logging of FOC parameters and fault states. To ensure data integrity, the Write Protect ($\overline{WP}$) and Hold ($\overline{HOLD}$) pins are pulled high via 10kΩ resistors (R9, R8). The Chip Select (CS_FLASH) is also pulled high via a 10kΩ resistor (R10) to prevent spurious writes during MCU boot-up.

- **Internal vs. External Encoder:** The board supports both the onboard MA732 magnetic encoder (U5) and a 6-pin JST-GH port (J4) for an external SPI encoder.
- **Hardware Arbitration:** The STM32 manages bus traffic using separate Chip Select pins. To switch between the onboard and external encoder, a physical solder jumper (JP1) is utilized on the MOSI line. The user must bridge JP1 with solder to use the internal MA732, or remove the solder bridge to utilize the external J4 port.

6.3 Debug and Programming Interface (SWD)

For firmware flashing and real-time debugging, a standard 6-pin SWD header (J5) is provided, exposing $3.3V$, GND, $SWDIO$, $SWCLK$, and the SWO trace output.

- **ESD Protection (U4):** Because industrial controllers are often handled in harsh environments, a TPD6E004RSER 6-channel ESD protection array safeguards the sensitive SWD and SPI lines from electrostatic discharge events.

6.4 Legacy PWM Control (Pixhawk Input)

For legacy integration (Mode I: ESC Mode), the board features a 6-pin JST-GH connector (J1) designed to read standard PWM/DShot signals from a flight controller without blocking the CPU.

- Channel 1 → PB4 (TIM3_CH1)
- Channel 2 → PB5 (TIM3_CH2)
- Channel 3 → PB10 (TIM2_CH3)
- Channel 4 → PB1 (TIM3_CH4)

6.5 I2C Expansion Interface

A dedicated 4-pin I2C header (J2) provides a 5V supply, GND, I2C_SCL, and I2C_SDA. This allows for the integration of external peripherals such as OLED status displays, temperature sensors, or secondary IMUs.

CHAPTER 7

Safety, Braking, and Fault Handling

7.1 Active Rheostatic Chopper (Braking System)

Regenerative braking on a 50.4V system can easily push the bus voltage past the 60V MOSFET limit. If V-Bus exceeds 55V, the STM32 triggers a dedicated Brake MOSFET, shorting the DC bus through an external high-wattage wirewound resistor (50W) connected to J-BRAKE.

7.2 Fault Tree Analysis and Hardware Protections

If a short circuit occurs, the DRV8353 physically cuts the gate drive in nanoseconds, pulls the nFAULT pin low, illuminates the Red Fault LED, and triggers an STM32 hardware interrupt to command a zero-torque safe state.

CHAPTER 8

Schematics and Layout

8.1 Hierarchical Schematic

The Indus-Drive 60V schematic is organized into a hierarchical structure consisting of four primary sheets to separate logical domains.

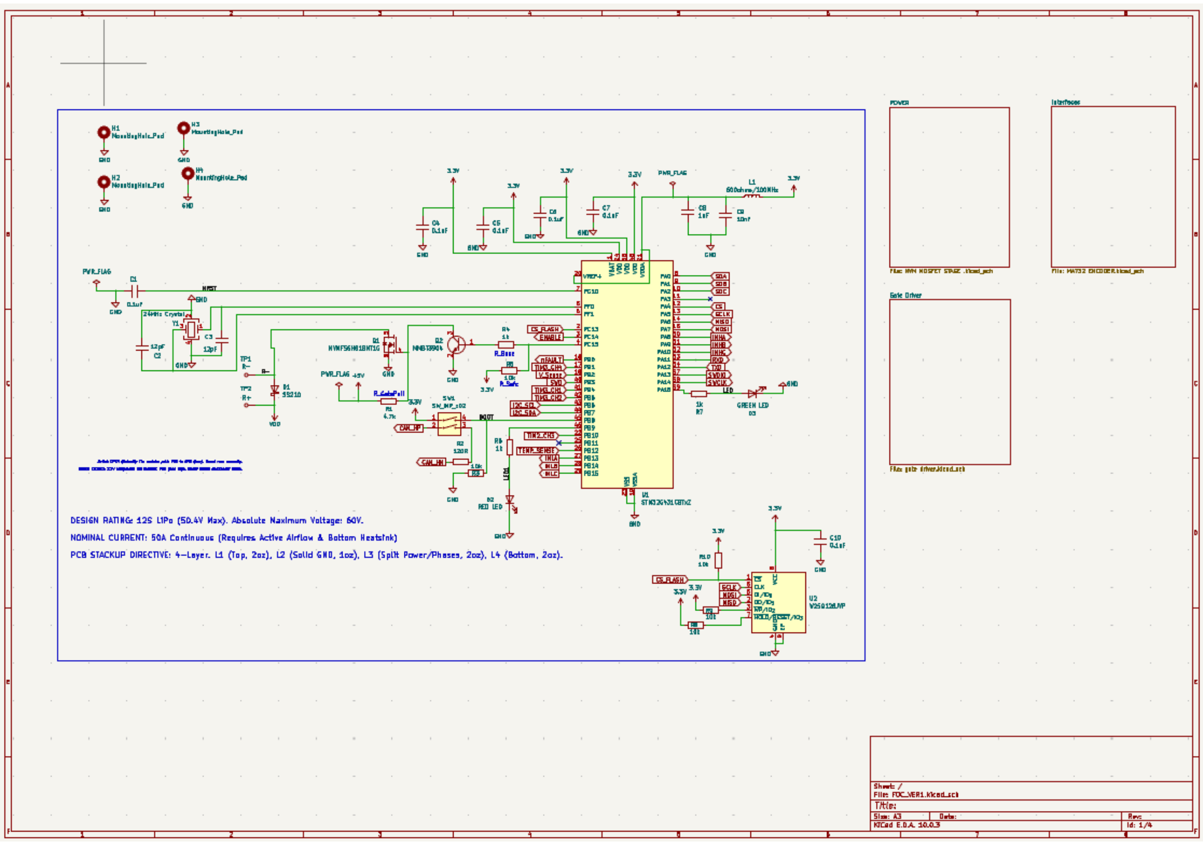

Figure 8.1: Schematic Page 1: Main Control (FOC_VER1)

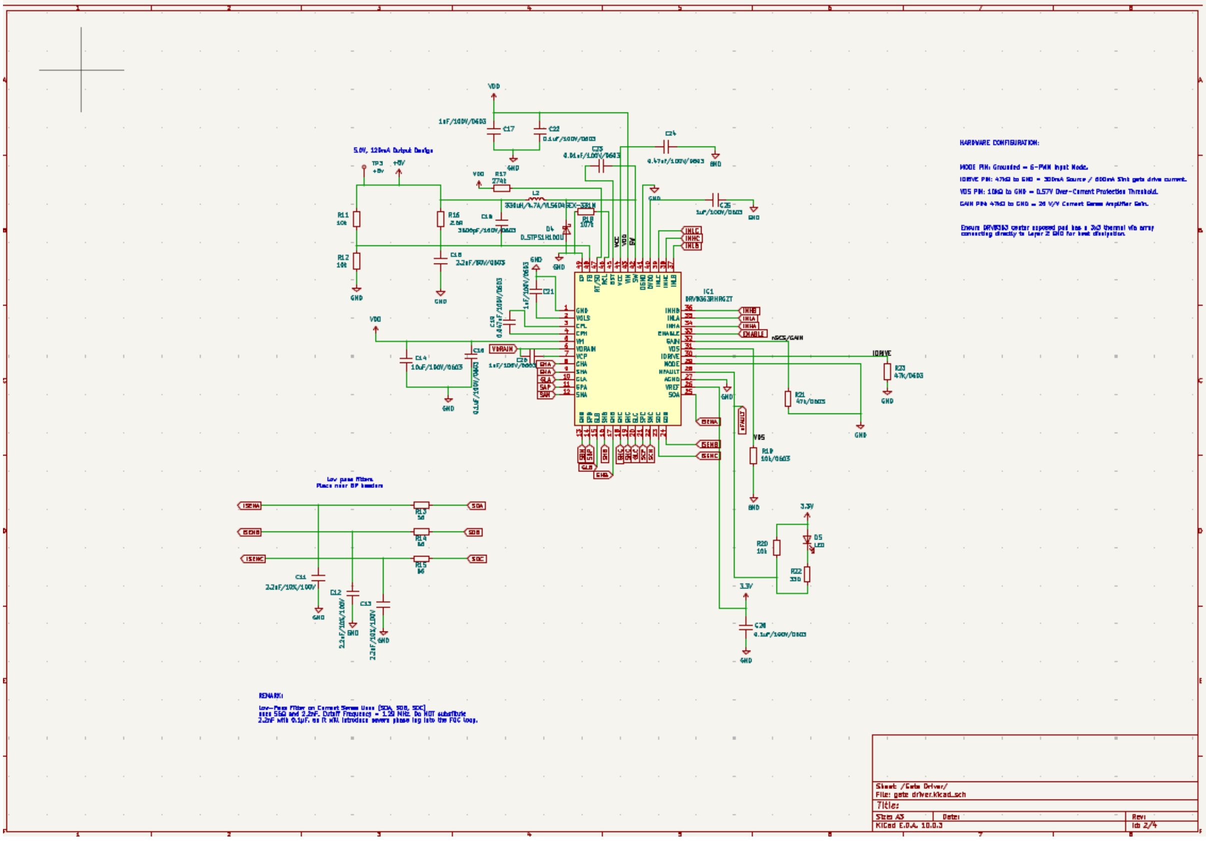

Figure 8.2: Schematic Page 2: Gate Driver

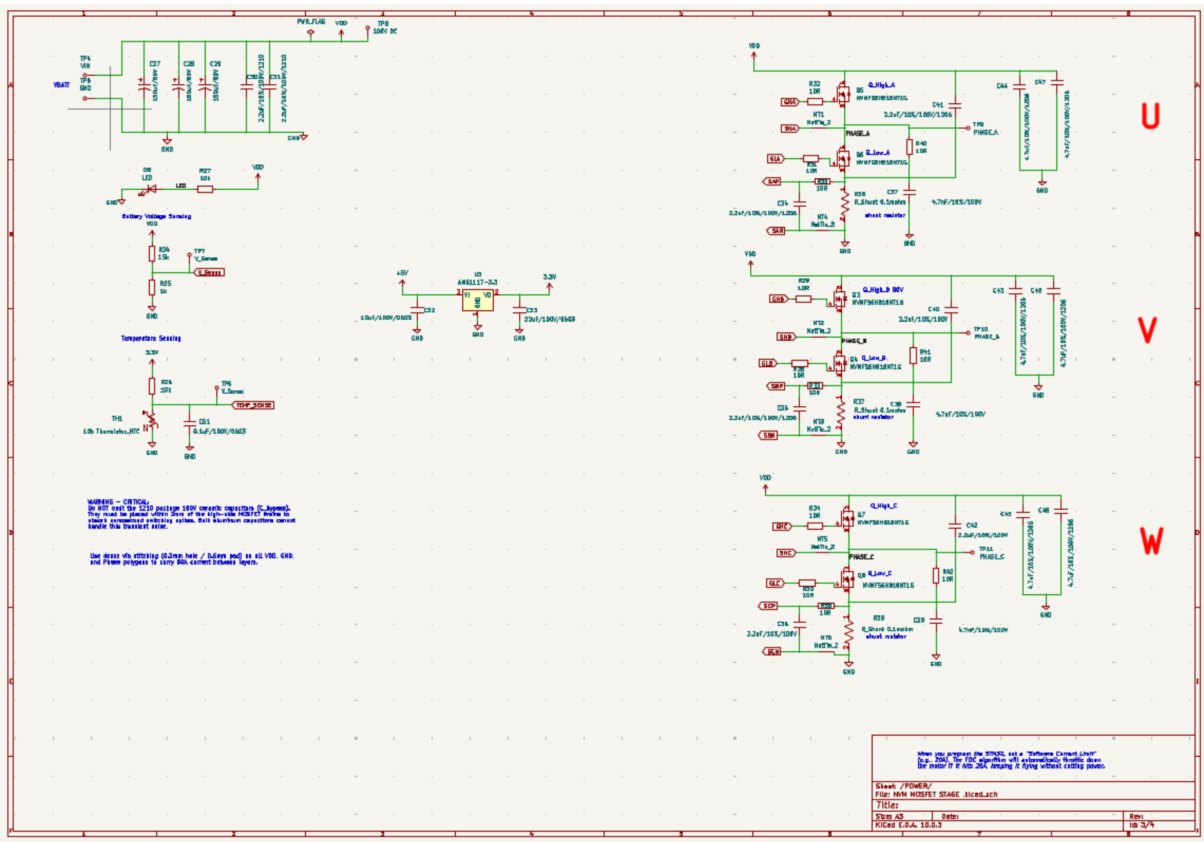

Figure 8.3: Schematic Page 3: POWER

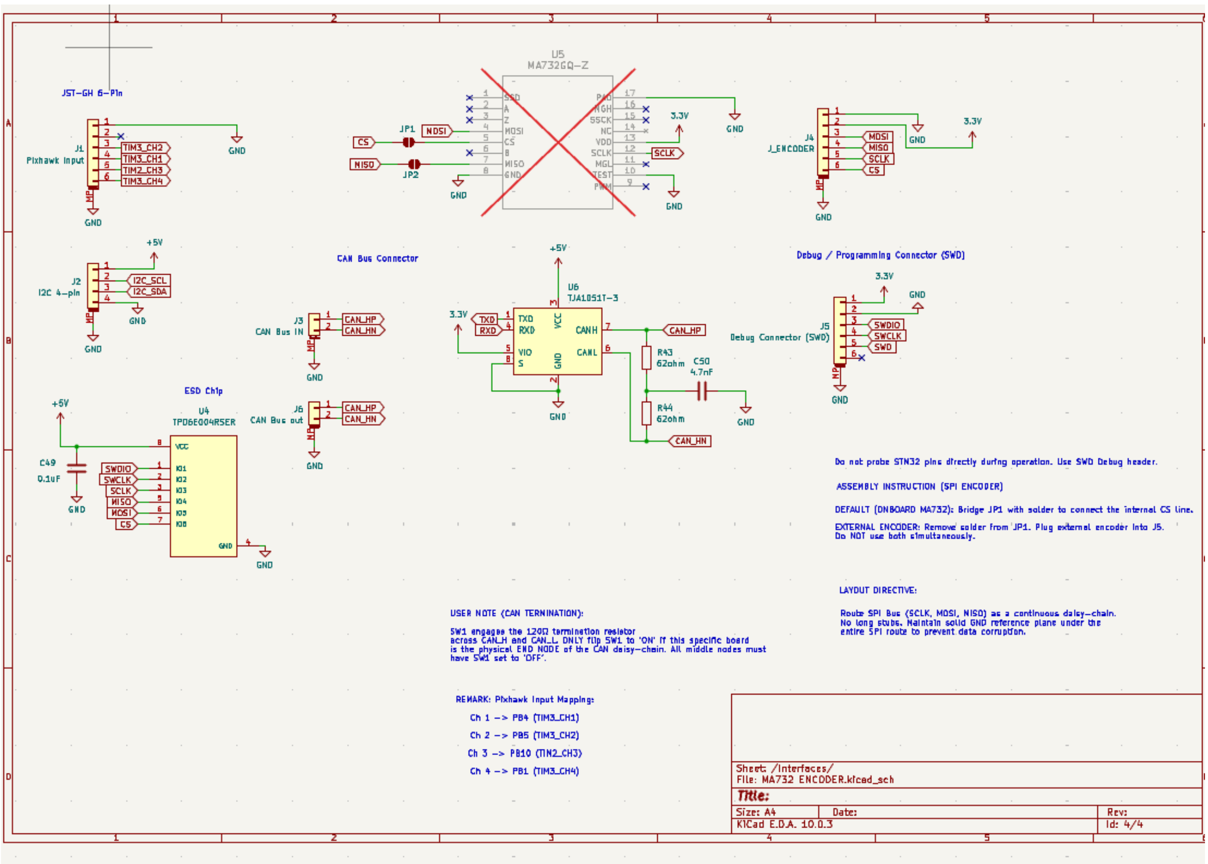

Figure 8.4: Schematic Page 4: Interfaces

8.2 PCB Layout Layers

The board utilizes a 4-layer 5cm $\times$ 5cm stackup to optimize thermal dissipation and isolate high-frequency digital signals from high-power switching nodes.

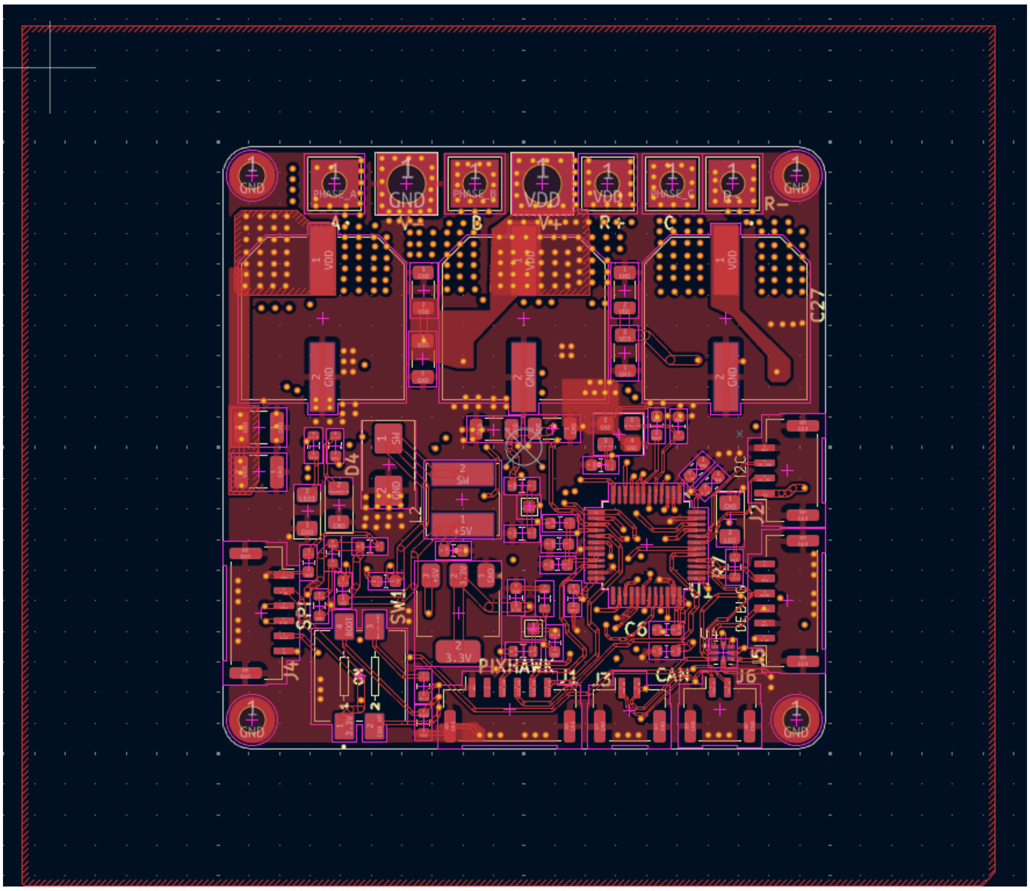

Figure 8.5: Layer 1: Top Signal and High Current Polygons

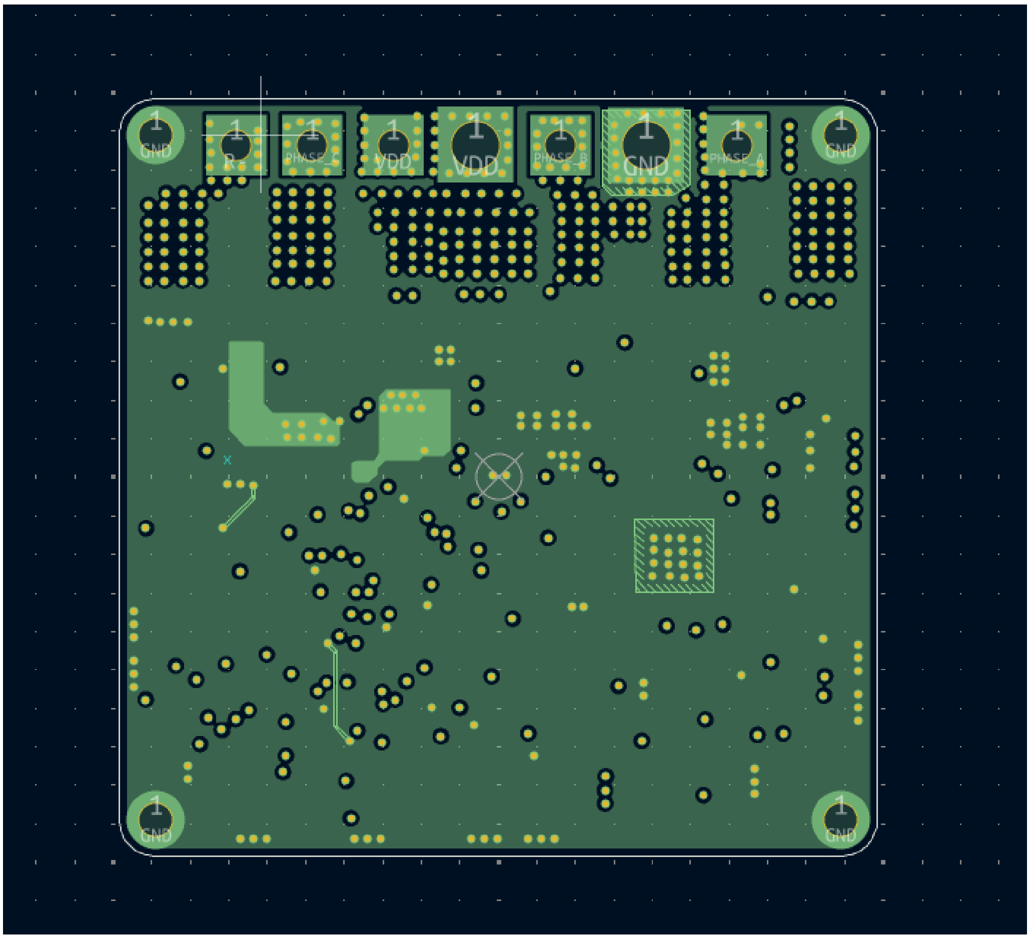

Figure 8.6: Layer 2: Solid Internal Ground (GND) Plane for EMI shielding

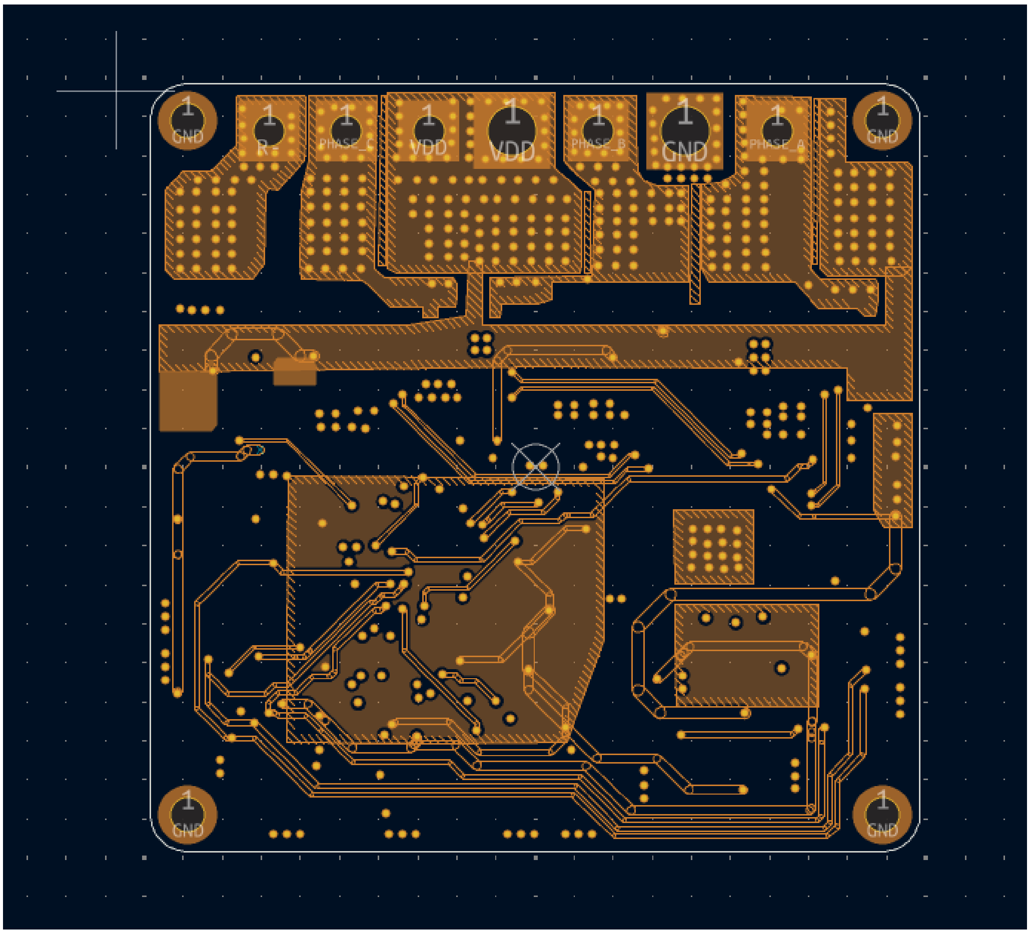

Figure 8.7: Layer 3: Internal Power (VCC) Plane

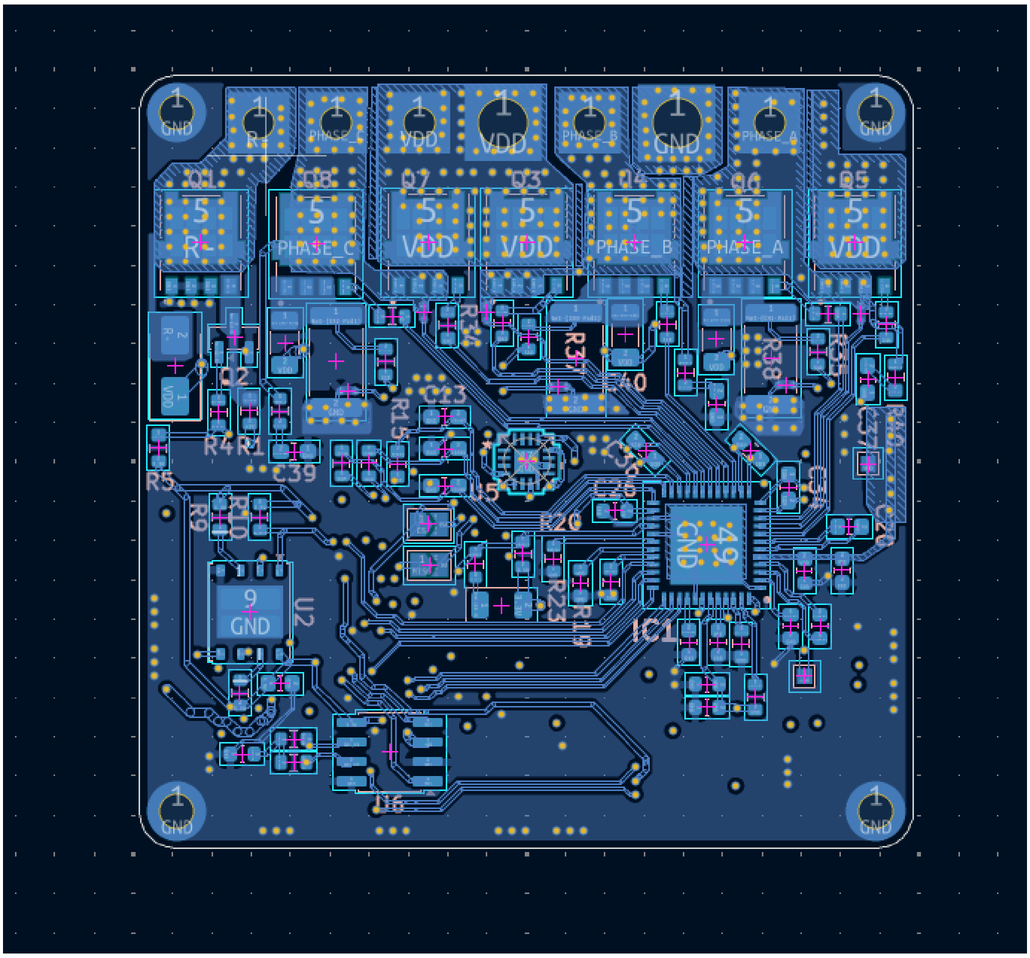

Figure 8.8: Layer 4: Bottom Signal and MA732 Quiet Island

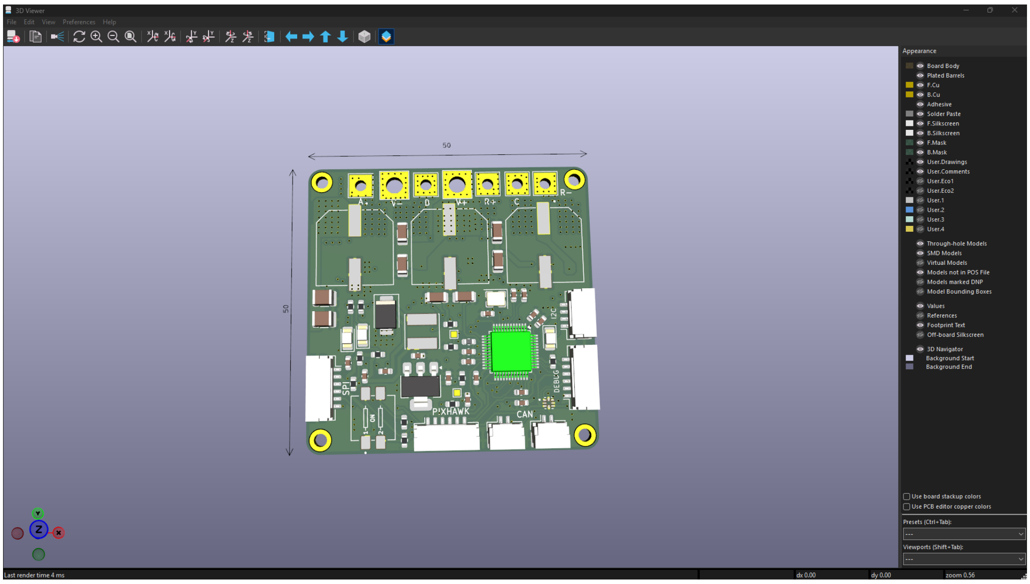

Figure 8.9: 3D Front view

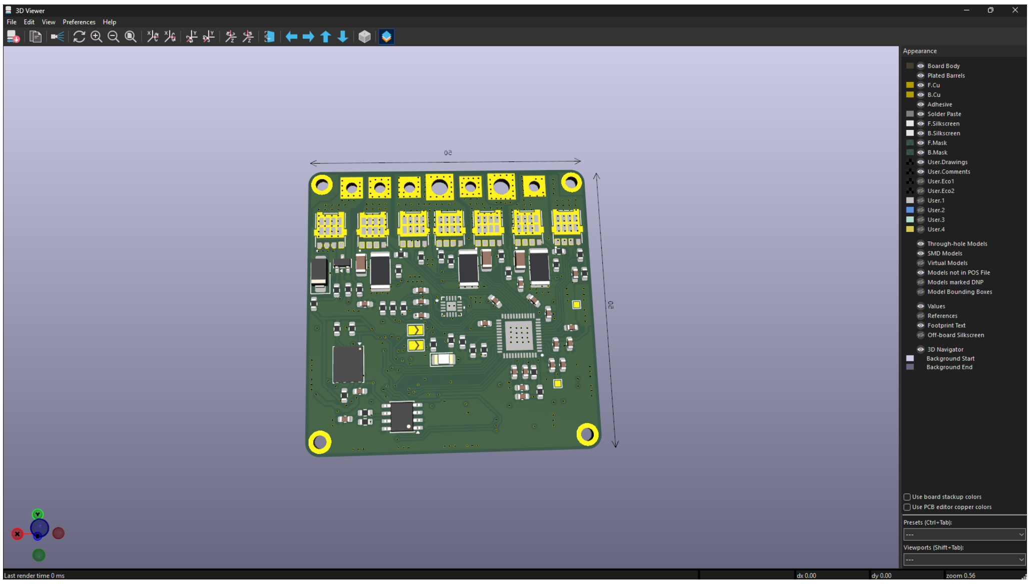

Figure 8.10: 3D Back view

CHAPTER 9

Project Repository

Source code, firmware, and design files are maintained in the official repository:

GitHub Repository Link

9.1 Reference Datasheets

- STM32G431 Microcontroller Datasheet: Download
- DRV8353 Gate Driver Datasheet: Download